



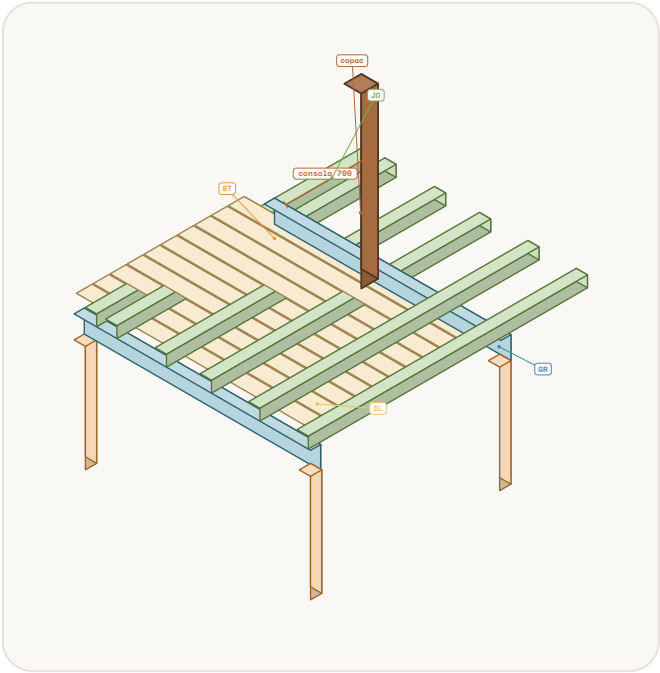
MANUAL DE CONSTRUCTIE · FAZA 1

Casuta din copac

Platforma pe patru stalpi, langa corcodusul din gradina —
pas cu pas, pentru Tudor si tata.

Ce construim

O casuta ridicata la 2,2 m pe patru stalpi (nu in copac — corcodusul e prea subtire ca s-o tina). In spate, langa gard, partea inchisa; in fata, un balcon deschis care iese 700 mm peste stalpi. Copacul trece prin podeaua balconului si il retezam ca masuta. **Numerotam stalpii: S1-S2 = spate (raman lungi, 4 m), S3-S4 = fata (se taie la +1872).**



Inaltime podea	2200 mm
Cadru stalpi	2100 × 1780 mm
Balcon (consola)	+700 mm
Grinzi	glulam 90×200
Joiste	6 × 100×100
Balustrada	1 m, goluri sub 9 cm
Copac	corcodus Ø160, masa

Cuprins

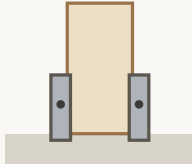
	Ce construim	2
	Materiale	4
	Scule	6
1	Stalpii in papuci	7
2	Nivel +2200 si taiere stalpi fata	10
3	Polite pe stalpii din spate	13
4	Grinda din spate pe polite	16
5	Grinda din fata pe varful stalpilor	19
6	Cele 6 joiste	22
7	Gaura copacului si masa	26
8	Blocaje intre joiste	29
9	Contravantuiri si scoatem proptelele	31
10	Dusumeaua de larice	34
11	Balustrada si poarta	37
	Siguranta & verificare	41

DE CE AI NEVOIE

Materialie

Toate piesele, cu codul lor (acelasi cod in toate desenele). Ce ai deja, ce e in comanda Hornbach si ce iei din offcut.

LE AI DEJA

			
ST 4	J0 4	PAP 4	F2 1
stalpi KVH 100x100	joiste (din lotul de 6)	papuci, deja in beton	grund Köber 4 L

IN COMANDA HORNBAACH (4.101 LEI)

			
GR 2	J0 2	DL 17	C1 4
grinzi glulam 90x200	completare joiste	dusumea larice	coltare 100x90
			
C2 12	H1 1 pac	H2 1 pac	H3 1 pac
coltare 90x65 joiste; +2 grinda spate = de luat	Heco 8x200	Heco 6x100	Heco 6x80
			
H4 2 pac	B1 1	B2 1 pac	B3 1 pac
inox 5x60	tija M12 (taie 4x ~220)	M12x120 baza	saibe M12

**B4****1 pac**

piulite M12

**F1****2**

ulei tec

DIN OFFCUT

**P0****2**

polite

**BL****~12**

blocaje

**RM****2**

traverse rama copac

**CF****6**

contrafise

**PT****~6**

proptele

FAZA 2 / DE ALES (NU OPRESTE STARTUL)

**SB****~7**stalpisor balustrada
58x58**SP****~24**

sipci 18x18, gol <9 cm

**MC****~7 m**

mana curenta

**BM****~12**

bulon M10 balustrada

**F3****1**

topcoat colorat exterior

Scule de care ai nevoie

Pregateste-le pe toate inainte de start. Pe fiecare fisa apar doar sculele etapei.

Boloboc / nivela cu furtun

verticalitate + orizontalitate

Bormasina + biti

suruburi Heco si gauri

Burghiu lemn 13 mm

gauri pentru buloane M12

Cheie 19 (M12)

buloane baza + polita

Cheie M10

balustrada (Faza 2)

Bomfaier

taiat tija filetata

Fierastrau / circular

taieri stalpi, dusumea, offcut

Echer mare

unghiuri drepte

Ruleta + creion

masuratori, trasaj

Cui 5 mm

distantier intre scanduri

Pensula

grund + ulei

2 persoane

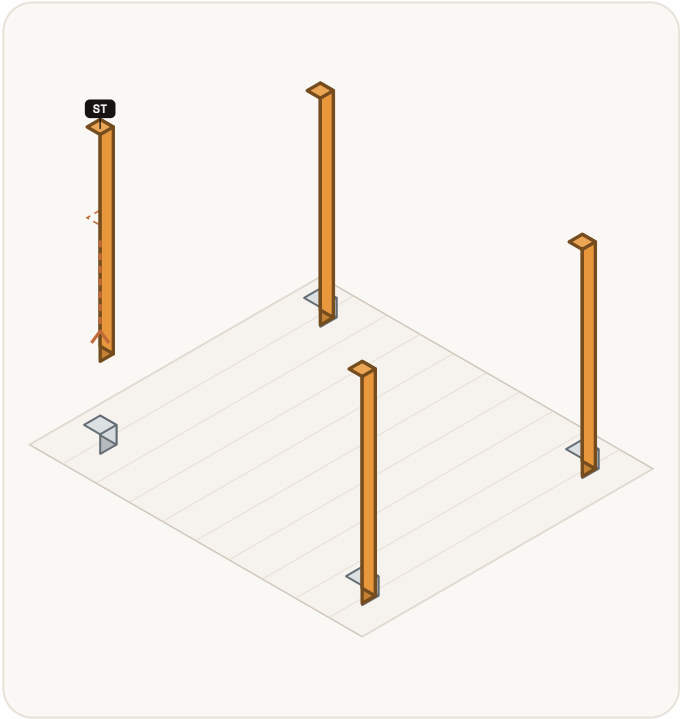
la grinzi si ridicari

Stalpii in papuci

~2-3 ore

Mediu

2 persoane



- ST stalp
- GR grinda
- JO joista
- DL dusumea
- PO polita
- C1/C2 coltar
- CF contrafisa

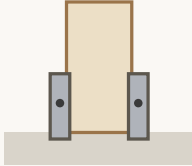
PIESE



ST

4

la lungime mare, NU taia



PAP

4

deja in beton



B2

8

2 / stalp



PT

~6

proptele

SCULE

Cheie 19 (M12)

Boloboc

Bormasina

2 persoane

PAS CU PAS

cap surub (fata)

ascuns (pe spate)

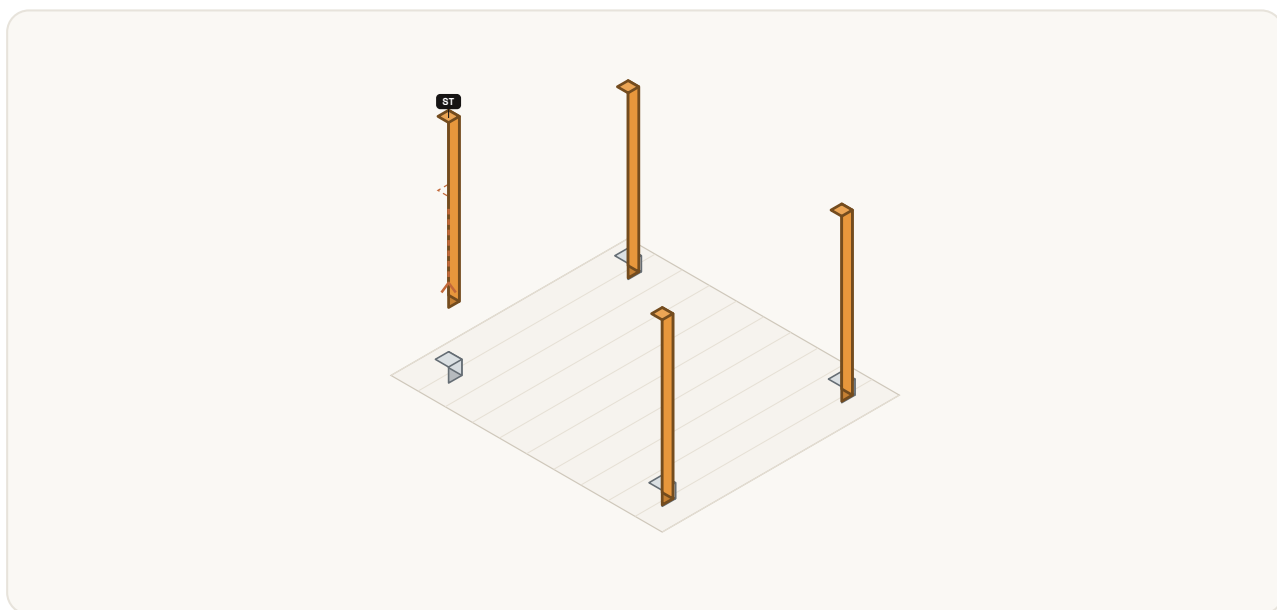
directie

oblic = la unghi

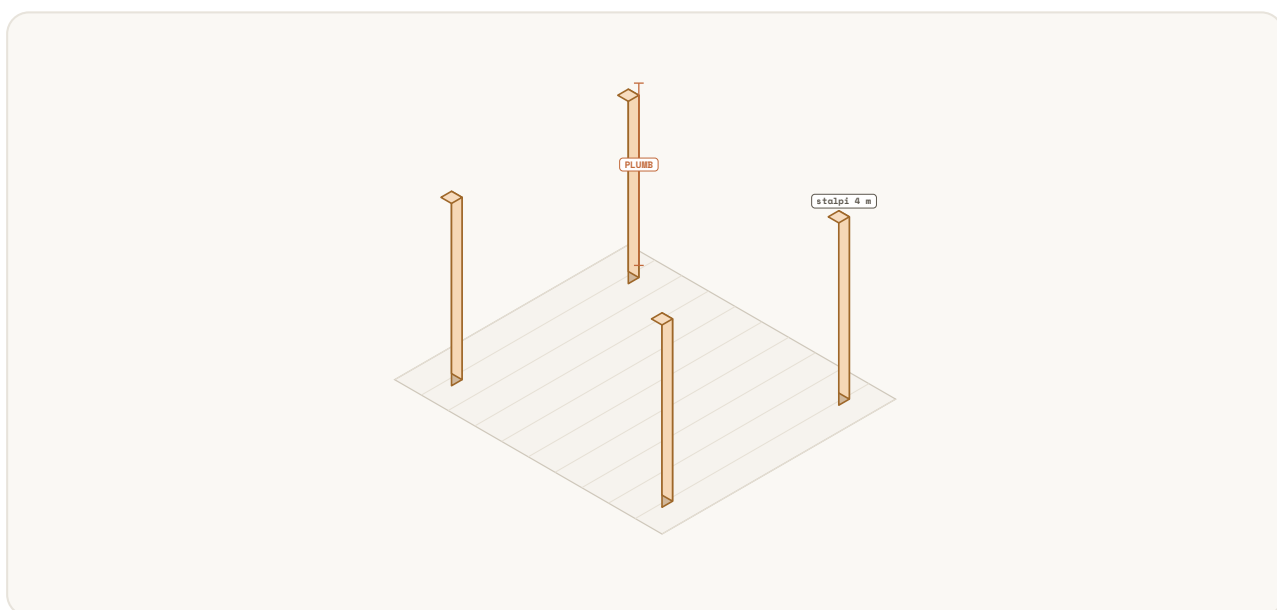
Folosim **suruburi + buloane**, nu cuie.

1

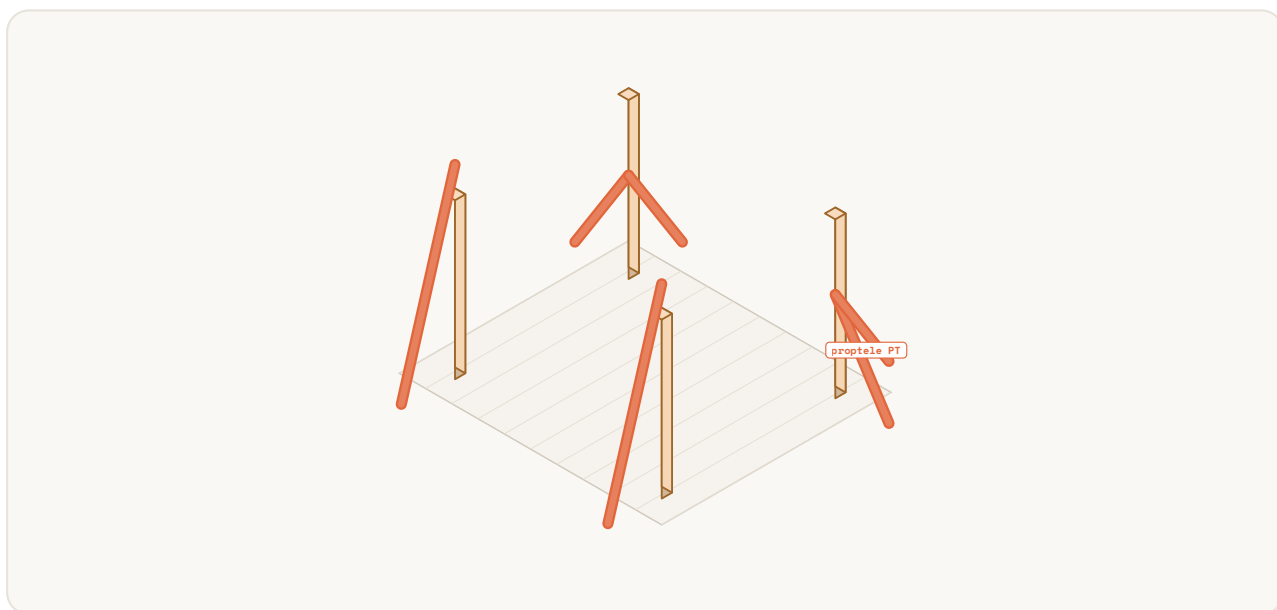
Aseaza toti 4 stalpii in papucii lor din beton, coborati drept in U. Cei 2 din spate raman LUNGI (4 m).

**2**

Adu fiecare stalp perfect vertical (la plumb), verificat pe doua fete cu bolobocul.



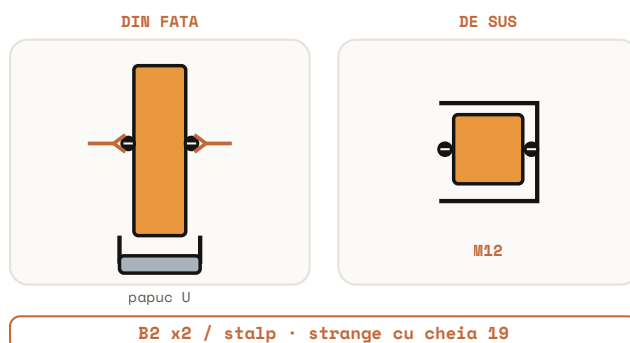
- 3 Sprijina fiecare stalp cu 2 proptele la baza. La cei 2 stalpi inalti din spate, pune si cate o proptea spre varf.



! CRITIC — opreste-te si verifica.

- 4 Prinde suruburile B2 in papuc DOAR slab (2 / stalp) — sa poti inca corecta. Le strangi la fisa 2.

ZOOM • CUM SE PRINDE



GATA CAND

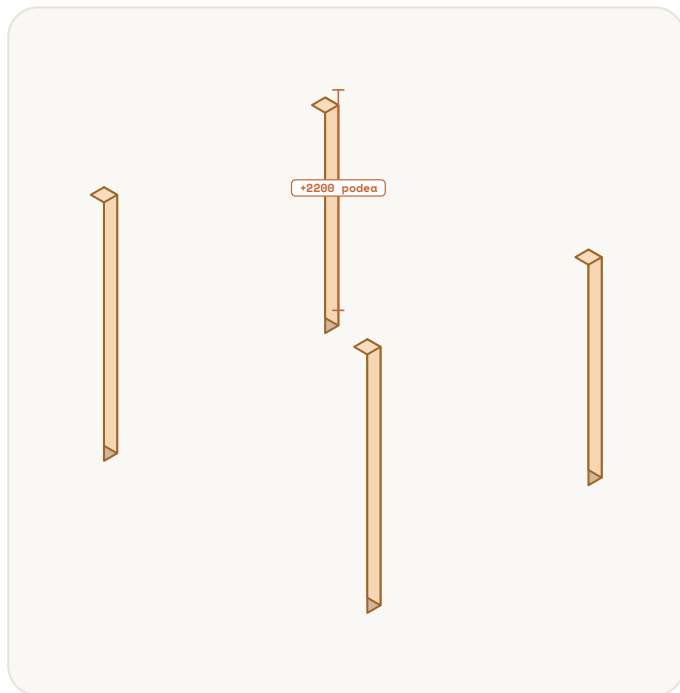
- ☒ Toti 4 stalpii verticali pe 2 fete
- ☒ Proptele la fiecare stalp
- ☒ Suruburi inca slabe

FISA 2 / 11

Nivel +2200 si taiere stalpi fata

~1-2 ore

CRITIC



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE



SCULE

Nivela cu furtun / laser

Creion

Fierastrau

Echer

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



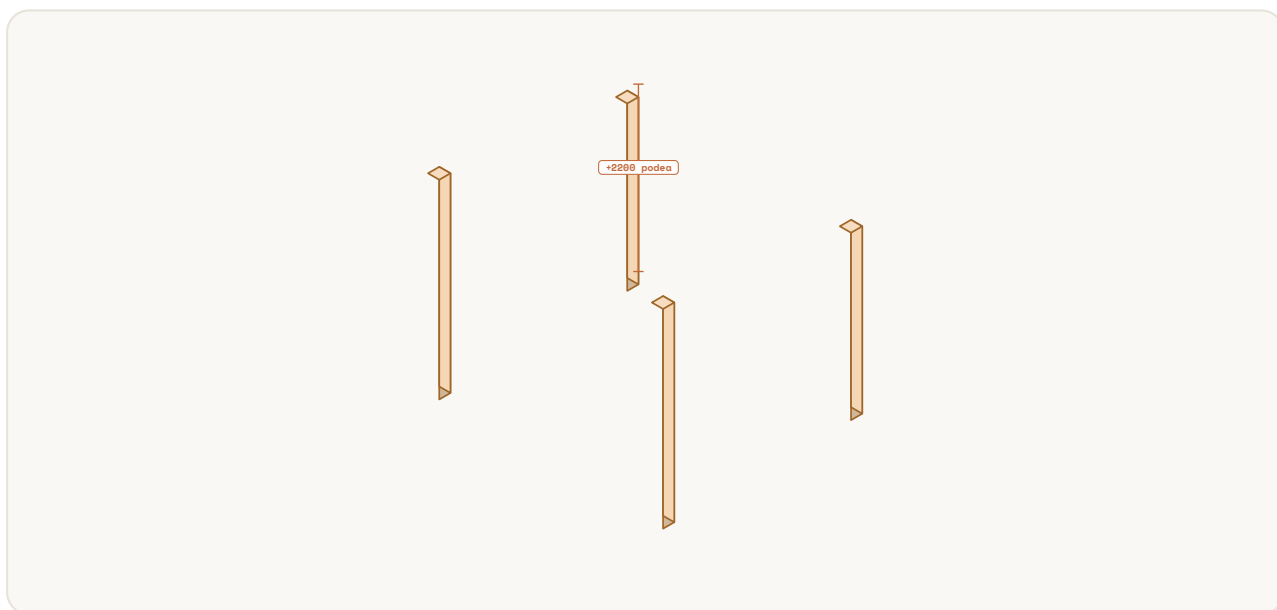
directie

oblic

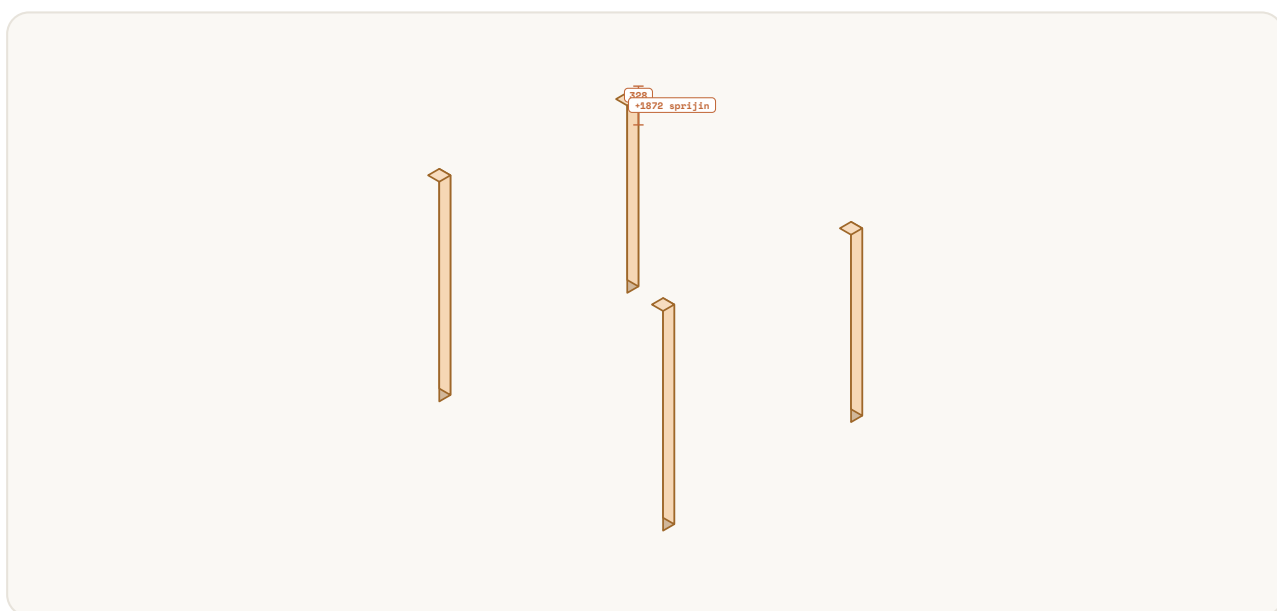
= la
unghi

Folosim **suruburi + buloane**, nu cuie.

- 1 Marcheaza +2200 (fata podelei) pe toti 4 stalpii. Foloseste o scandura dreapta + boloboc, nu masura separat.

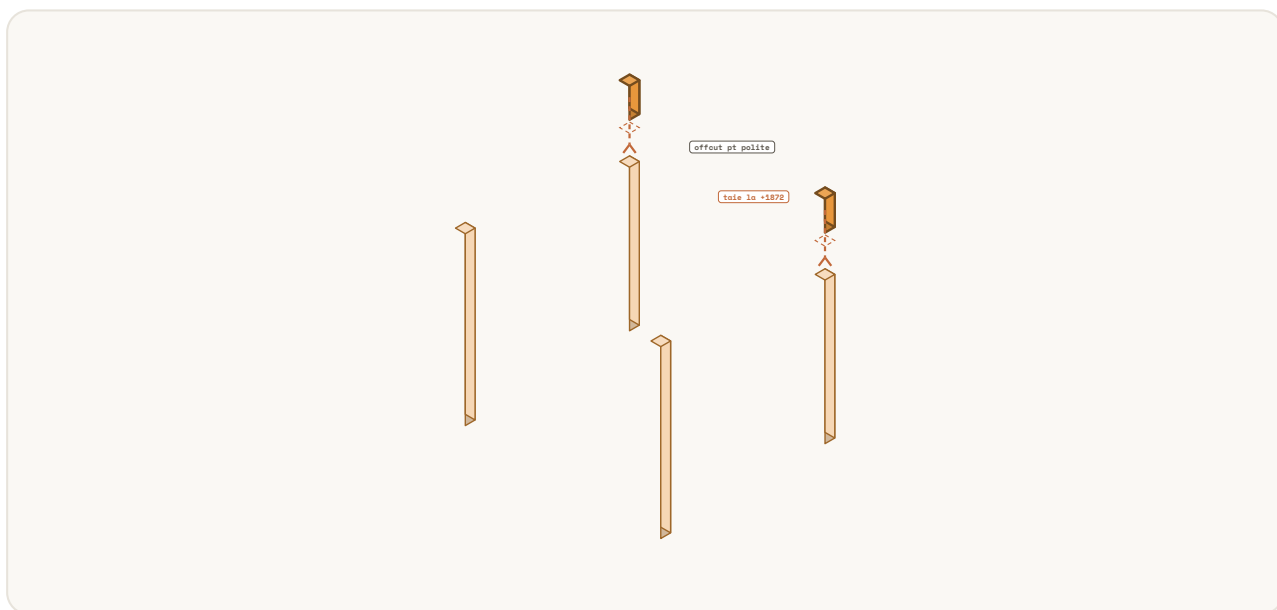


- 2 Coboara 328 mm si marcheaza +1872 — aici sta talpa GRINZII pe polite. Grinda 200 + joista 100 + dusumea 28 = 328, asa ajungi la +2200, fata podelei.



! CRITIC — opreste-te si verifica.

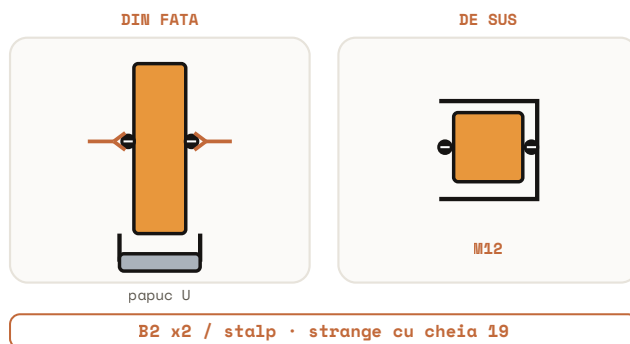
- 3 Taie DOAR stalpii din fata (S3, S4) la +1872. Partea taiata o pastrezi pentru polite. Stalpii din spate raman INTREGI.



! CRITIC — opreste-te si verifica.

- 4 Teseste muchia taiata. Abia acum strange definitiv suruburile B2 de la baza.

ZOOM • CUM SE PRINDE



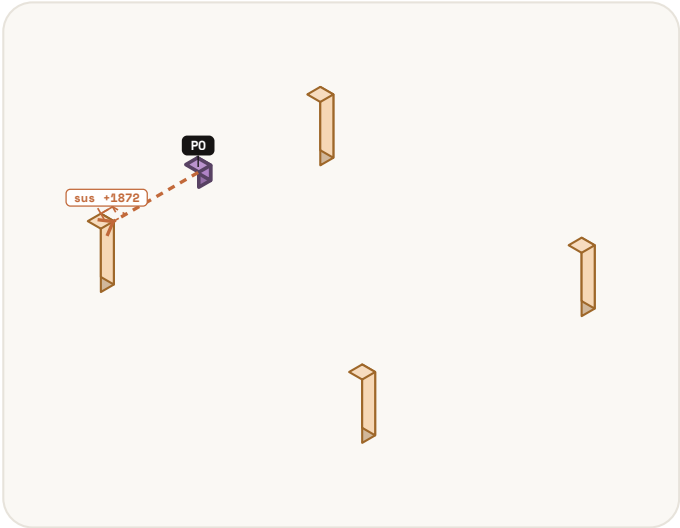
GATA CAND

- ☒ Linia +1872 identica pe toti
- ☒ Doar stalpii fata taiati
- ☒ Stalpii spate INTREGI
- ☒ Suruburi baza stranse

Polite pe stalpii din spate

~1 ora

CRITIC



ST stalp **GR** grinda **JO** joista **DL** dusumea **PO** polita **C1/C2** coltar **CF** contrafisa

PIESE

P0	B1	B3	B4
2	4	8	4
offcut 100x100	tija ~220	2 / bulon: cap + piulita	piulite

SCULE

Bormasina + burghiu 13 Cheie 19 Bomfaier

PAS CU PAS

● cap surub (fata)

○ ascuns (pe spate)

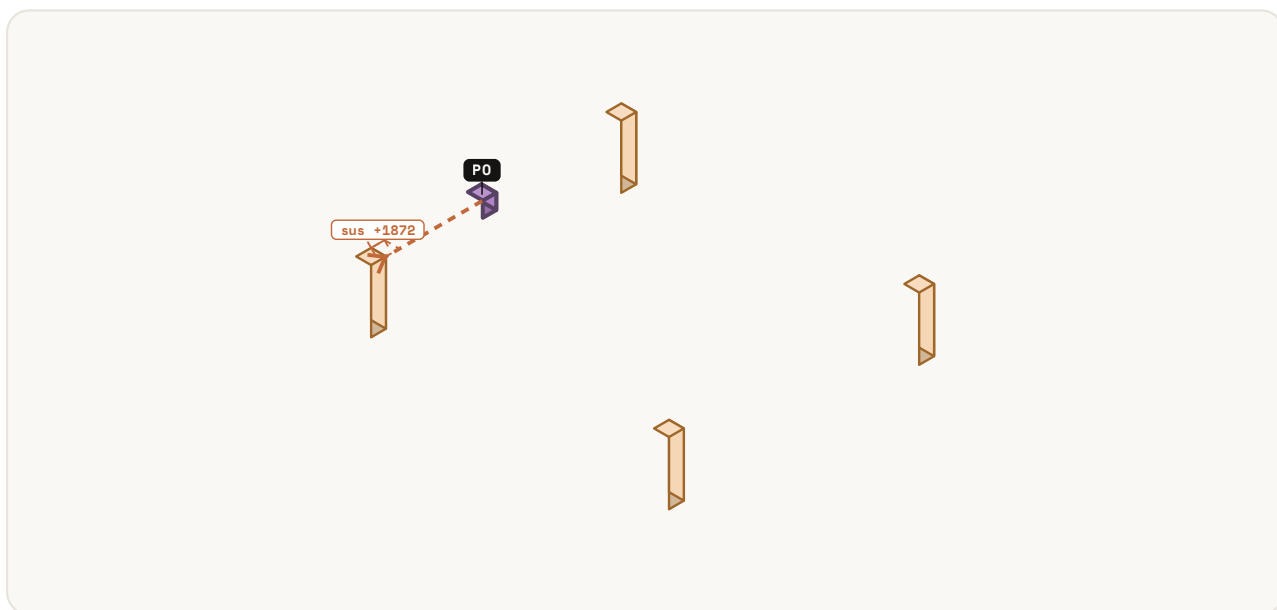
→ directie

oblic = la unghi

Folosim **suruburi + buloane**, nu cuie.

1

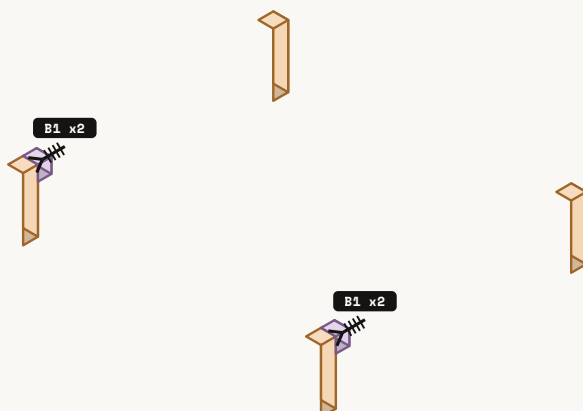
Aseaza polita (bloc 100x100 din offcut) pe fata interioara a stalpului din spate, cu fata de sus la +1872.

**2**

Gaureste prin polita in stalp: 2 gauri de 13 mm.

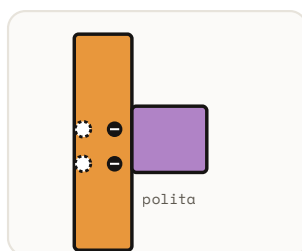
3

Bate 2 buloane M12 (B1) cu saiba (B3) prin polita in stalp; strange cu piulita (B4) pe spate. Repeta la al doilea stalp.

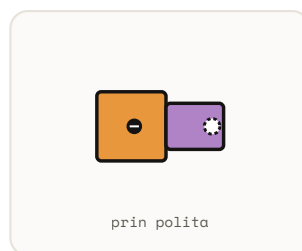


ZOOM · CUM SE PRINDE

LATERAL



DE SUS



B1 x2 / polita · cap pe fata, piulita pe spate

! CRITIC — opreste-te si verifica.

GATA CAND

-  Ambele polite sus la +1872
-  2 buloane stranse / polita
-  Polita nu se misca

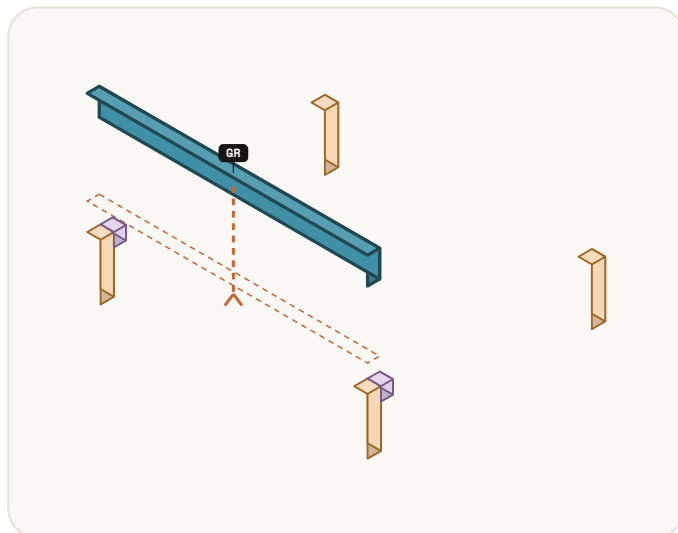
FISA 4 / 11

Grinda din spate pe polite

~1 ora

CRITIC

2 persoane



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE

GR	C2	H1
grinda spate	coltar anti-smulgere	3 oblice / capat

SCULE

2 persoane Bormasina Nivela

PAS CU PAS

● cap surub
(fata)

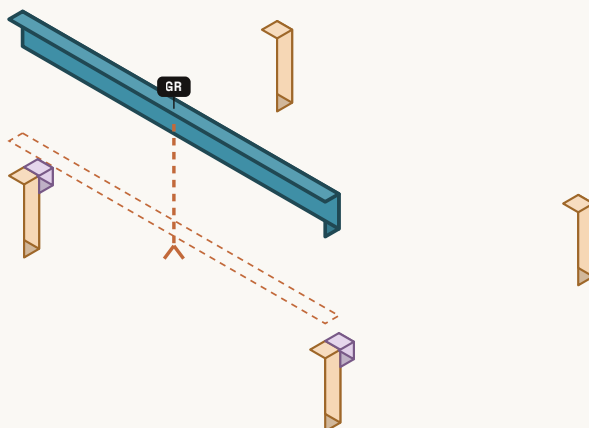
○ ascuns (pe
spate)

→ directie **oblic** = la
unghi

Folosim **suruburi + buloane**, nu cuie.

1

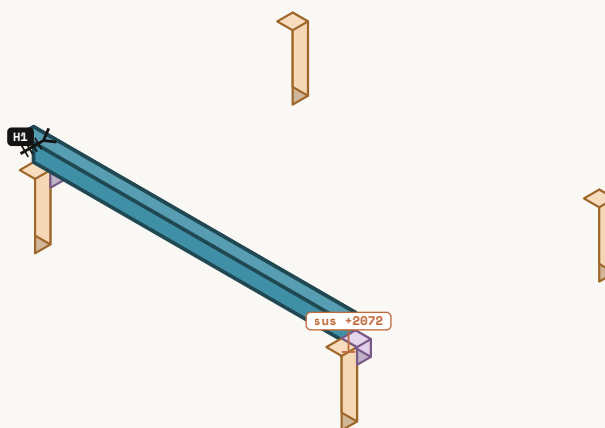
Ridicati in DOI grinda din spate si asezati-o pe cele doua polite. Sta pe polita — nu o tineti voi.



! CRITIC — opreste-te si verifica.

2

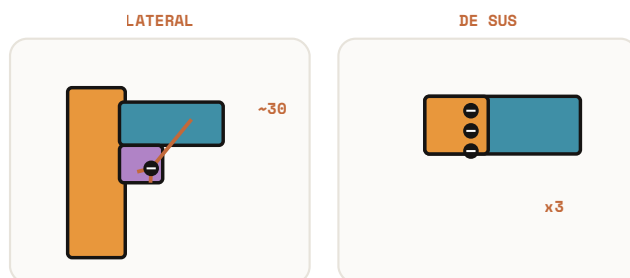
Verifica fata de sus la +2072 si orizontalitatea pe toata lungimea.



3

Prinde grinda de fiecare stalp cu 3 suruburi H1 oblice PLUS un coltar metalic in stalp si in grinda (anti-smulgere). Grinda nu trebuie sa se poata ridica de pe polita.

ZOOM · CUM SE PRINDE



H1 oblic in stalp · x3 · tine grinda lipita

! CRITIC — opreste-te si verifica.

GATA CAND

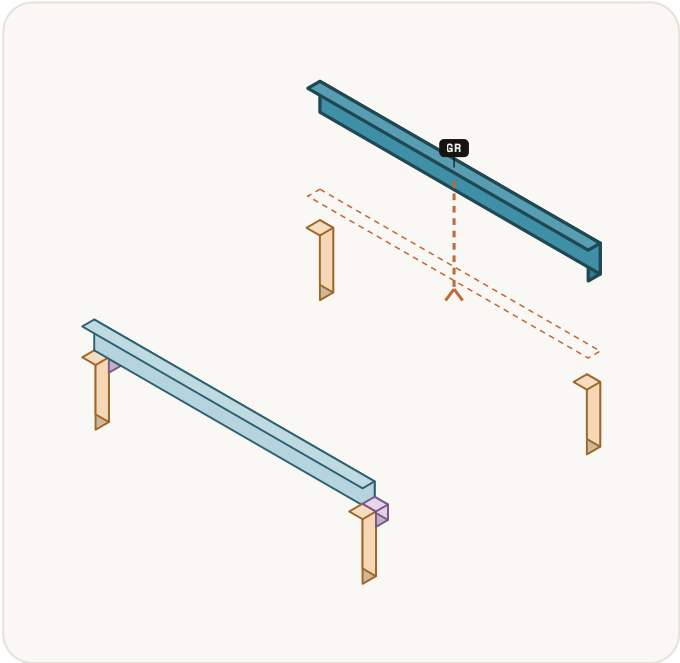
- ☒ Grinda pe ambele polite
- ☒ Fata sus la +2072
- ☒ Coltar anti-smulgere la fiecare capat
- ☒ Buloane polita REVERIFICATE dupa asezarea grinzii

Grinda din fata pe varful stalpilor

~1 ora

CRITIC

2 persoane



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE

GR1grinda fata

C142 / stalp

H1~12in coltare

SCULE

2 persoane

Bormasina

Nivela

PAS CU PAS

•

cap surub (fata)

◉

ascuns (pe spate)

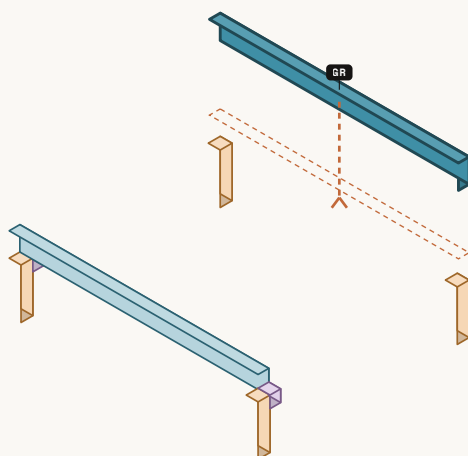
→

directie oblic = la unghi

Folosim **suruburi + buloane**, nu cuie.

1

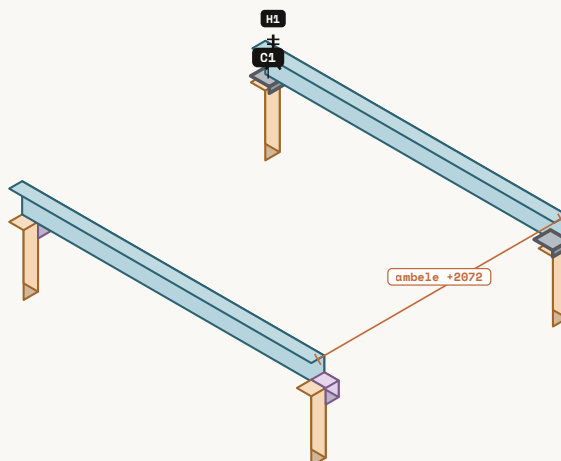
Asezati grinda din fata PE varful stalpilor S3/S4 (taiati). Sprijin direct pe lemn.



! CRITIC — opreste-te si verifica.

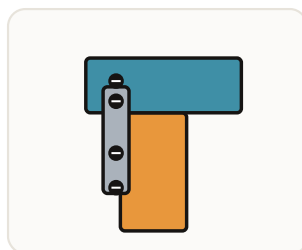
2

Prinde cu cate un coltar C1 pe fiecare fata (2/stalp) + suruburi H1 in stalp si in grinda.

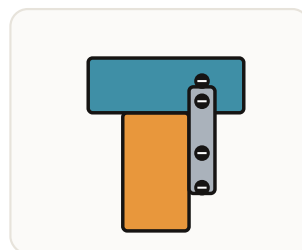


ZOOM · CUM SE PRINDE

FATA A



FATA B (opus)



C1: 2 coltare / stalp · fiecare cu 2 in stalp + 2 in grinda

3

Verifica: ambele grinzi horizontale, varful la +2072. Ai cadrul de baza.

GATA CAND

- ☒ Grinda pe varful stalpilor
- ☒ 2 coltare C1 / stalp
- ☒ Ambele grinzi la +2072

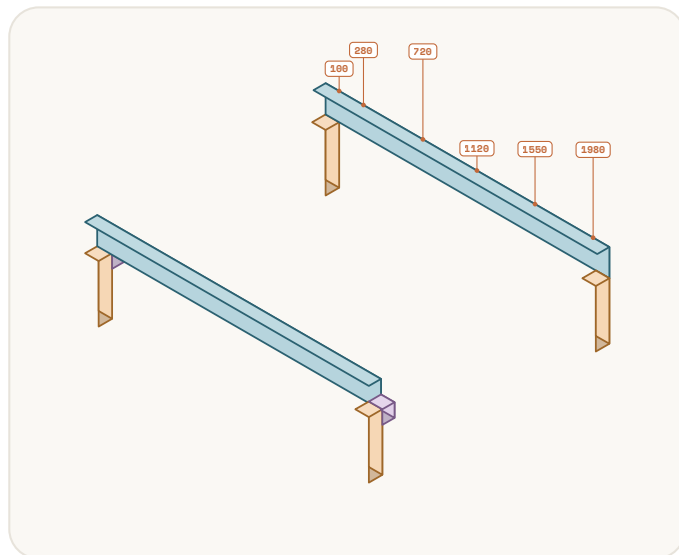
FISA 6 / 11

Cele 6 joiste

~2 ore

CRITIC

2 persoane



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE



SCULE

Bormasina

Ruleta

Creion

Echer

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

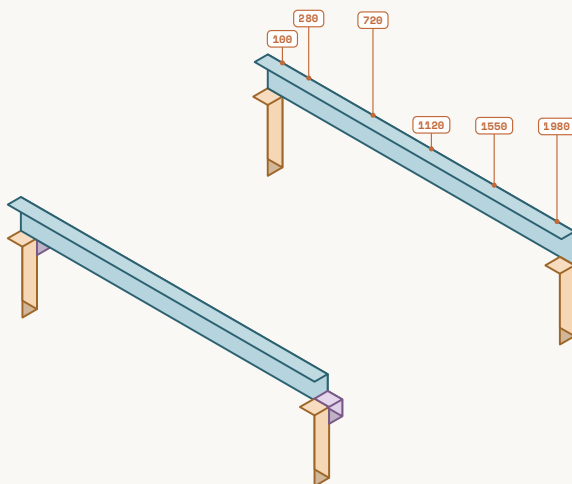
oblic

= la
unghi

Folosim **suruburi + buloane**, nu cuie.

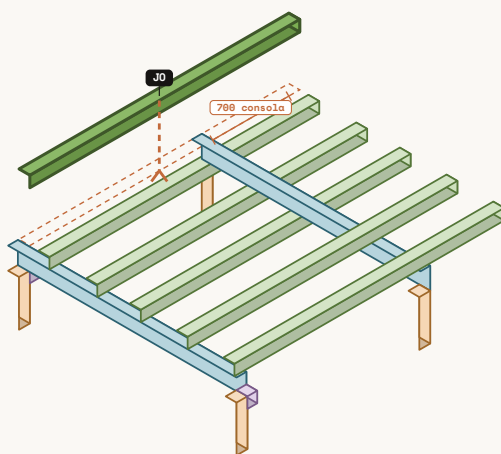
1

Marcheaza pe ambele grinzi pozitiile celor 6 joiste, de la coltul S4: 100 / 280 / 720 / 1120 / 1550 / 1980 mm.



2

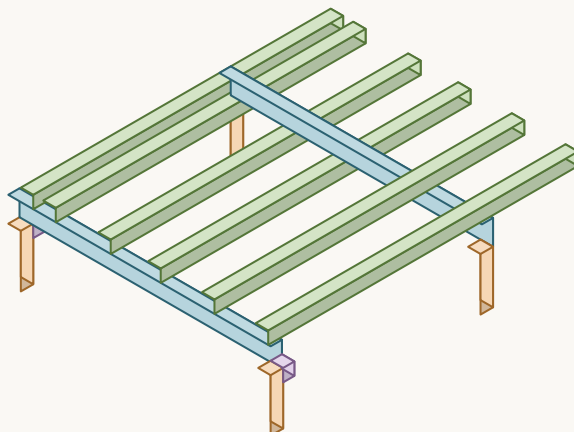
Aseaza prima joista peste ambele grinzi. Capatul din fata iese 700 mm in gol (consola = balconul).



! CRITIC — opreste-te si verifica.

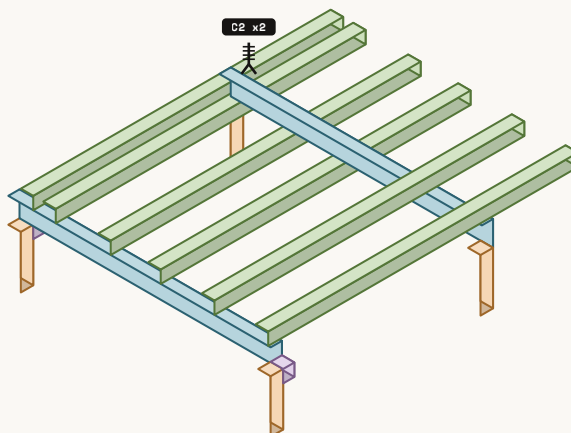
3

Aseaza toate cele 6 joiste la pozitiile marcate.



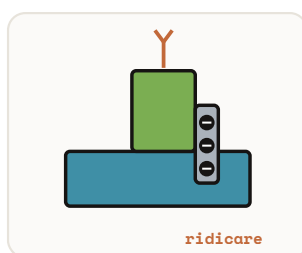
4

La FIECARE reazem (fata + spate) prinde un coltar C2 pe lateral — 12 in total. Tine joista jos la consola.

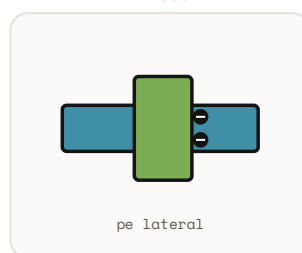


ZOOM · CUM SE PRINDE

LATERAL



DE SUS



C2 x2 / joista (fata + spate) · TOTAL 12

! CRITIC — opreste-te si verifica.

GATA CAND

6 joiste la pozitii

Consola 700 mm

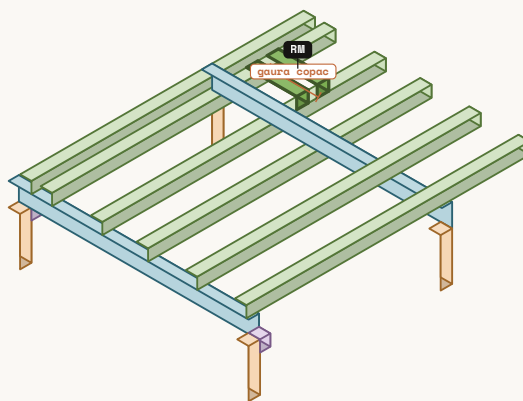
12 coltare C2

FISA 7 / 11

Gaura copacului si masa

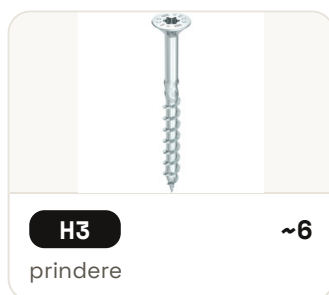
~1-2 ore

Atentie



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE



SCULE

Fierastrau

Bormasina

Ruleta

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



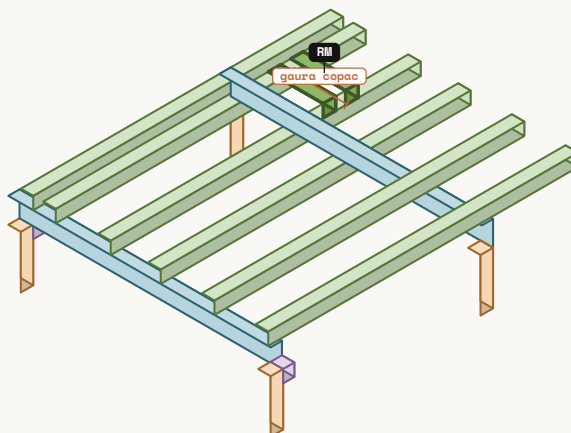
directie

oblic

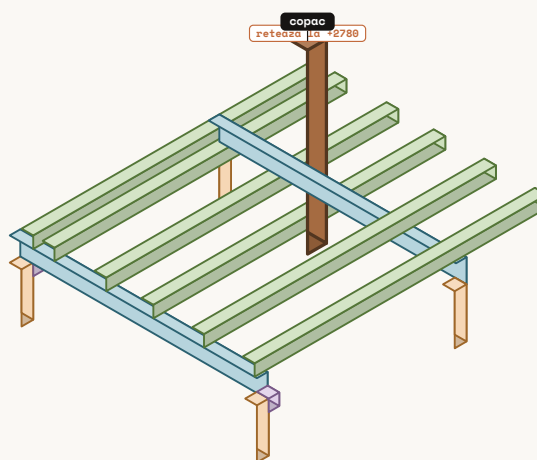
= la
unghi

Folosim **suruburi + buloane**, nu cuie.

- 1 Intre joistele de la 280 si 720 — trunchiul cade chiar intre ele, fara sa le atinga — monteaza 2 traverse scurte (RM) care inchid rama gaurii.



- 2 Reteaza corcodusul la ~+2780 de la sol (~580 peste podea). Lasa joc de 3-5 cm jur-impjurul trunchiului.



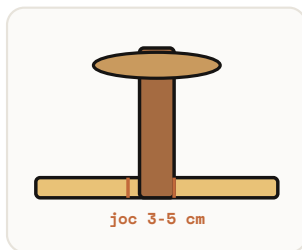
! CRITIC — opreste-te si verifica.

3

Blatul mesei se fixeaza de podea, prin cadrul lui — NICIODATA de copac.

ZOOM · CUM SE PRINDE

LATERAL



REGULA



Blatul pe cadrul lui propriu · NICIODATA pe copac

! CRITIC — opreste-te si verifica.

GATA CAND

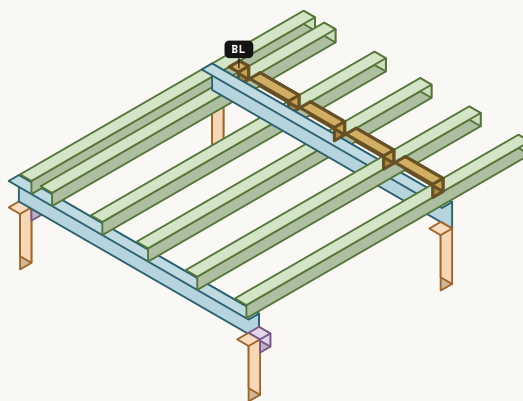
-  Rama intre joistele 280-720
-  Joc 3-5 cm in jurul trunchiului
-  Blatul NU pe copac

FISA 8 / 11

Blocaje intre joiste

~1 ora

Normal



ST stalp **GR** grinda **JO** joista **DL** dusumea **PO** polita **C1/C2** coltar **CF** contrafisa

PIESE



BL

offcut

~10



H3

oblic

~20

SCULE

Fierastrau

Bormasina

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

oblic

= la
unghi

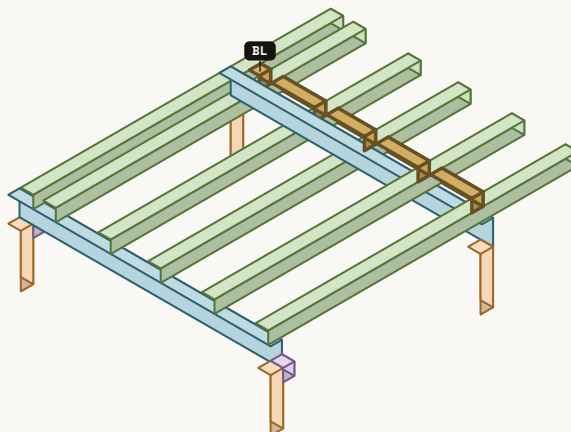
Folosim **suruburi + buloane**, nu cuie.

1

Taie bucati scurte de lemn (blocaje, BL) cat distanta dintre joiste.

2

Pune cate un blocaj intre joiste, PESTE fiecare grinda (fata si spate).

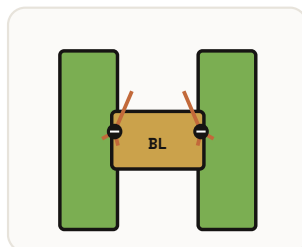


3

Prinde fiecare blocaj cu suruburi H3 oblic. Opresc joistele sa se răsuceasca — podeaua devine ferma.

ZOOM · CUM SE PRINDE

LATERAL



DE SUS



H3 oblic · cate 1-2 / capat

GATA CAND

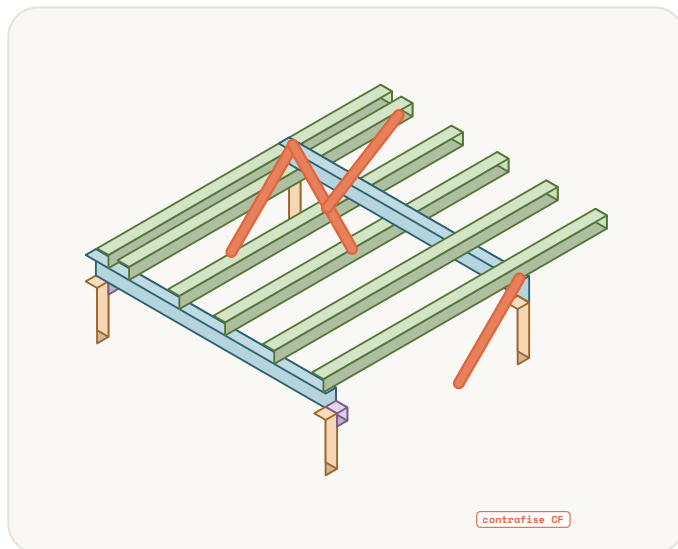
-  Blocaje peste ambele grinzi
-  Prinse oblic cu H3
-  Podeaua e ferma

FISA 9 / 11

Contravantuiri si scoatem proptelele

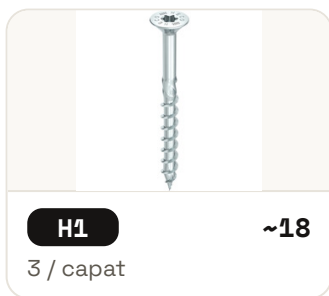
~2 ore

CRITIC



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE



SCULE

Fierastrau

Bormasina

Echer

PAS CU PAS

● cap surub
(fata)

○ ascuns (pe
spate)

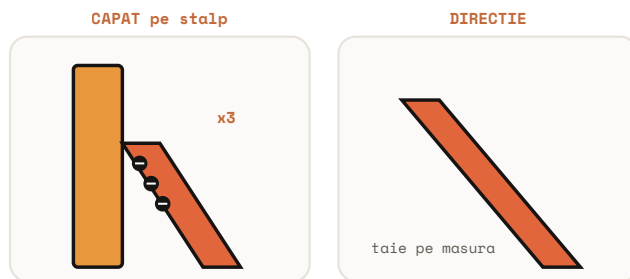
→ directie oblic = la
unghi

Folosim **suruburi + buloane**, nu cuie.

1

Masoara si taie capatul fiecarei contrafise (diagonala) pe potriveala.

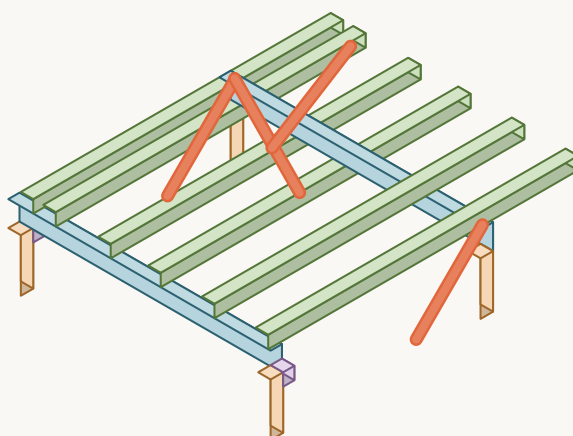
ZOOM · CUM SE PRINDE



H1 x3 / fiecare capat · fara coltar

2

Monteaza contravantuirile in planul stang, drept, fata si sub nasul consolei. La contrafisa de sub nas prinde-o pe LATERAL sau adauga un coltar — acolo lucreaza si la intindere. Platforma devine rigida.



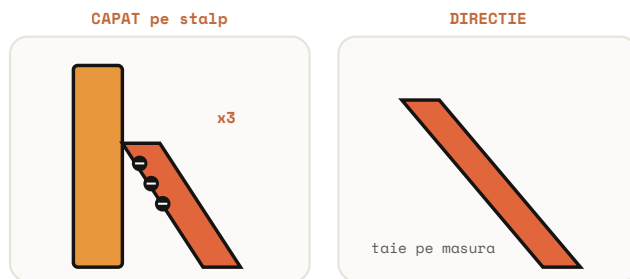
contrafise CF

! CRITIC — opreste-te si verifica.

3

Prinde fiecare capat cu 3 suruburi H1.

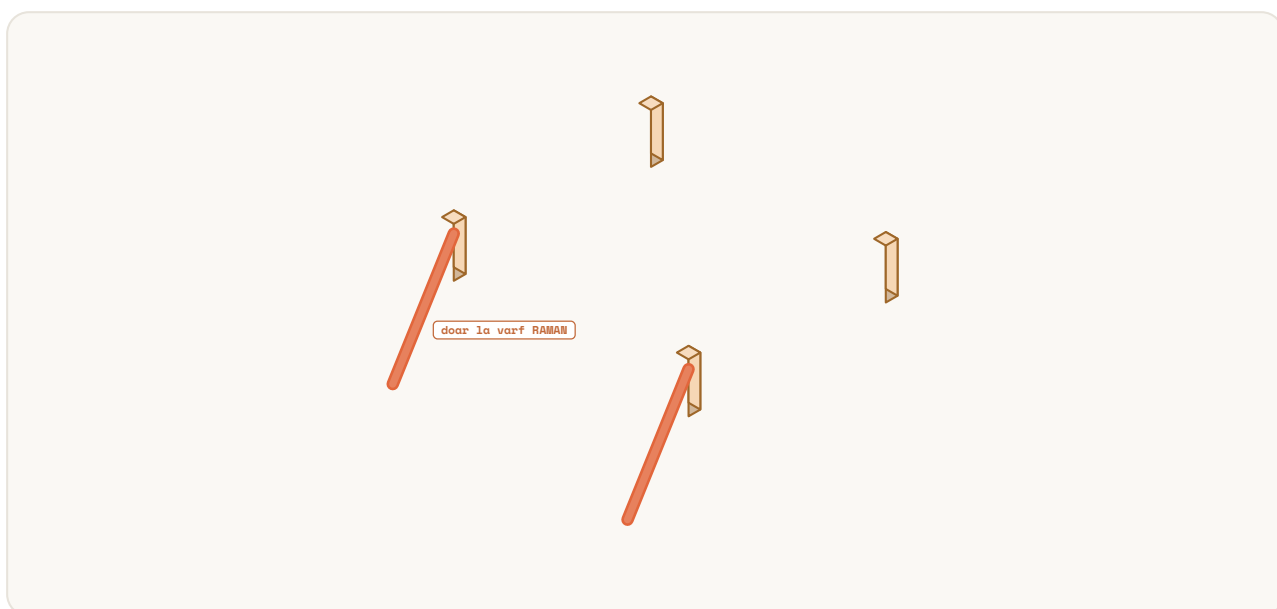
ZOOM · CUM SE PRINDE



H1 x3 / fiecare capat · fara coltar

4

SCOATE proptelele de la baza. Cele de la varful stalpilor inalti din spate RAMAN pana la peretii din Faza 2.



! CRITIC — opreste-te si verifica.

GATA CAND

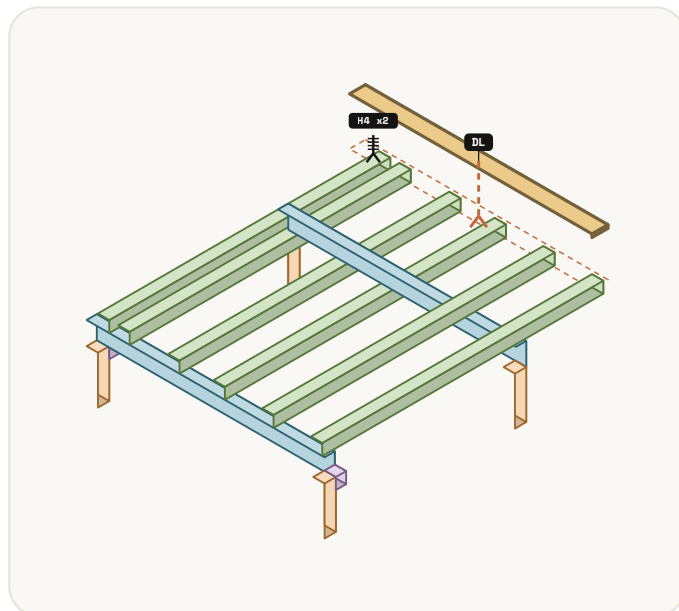
- ☒ Diagonale pe toate planurile + sub consola
- ☒ 3 suruburi / capat
- ☒ Proptele baza scoase; varf raman

FISA 10 / 11

Dusumeaua de larice

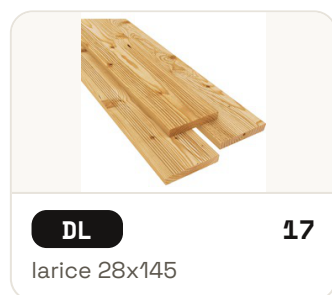
~3-4 ore

Normal



ST stalp **GR** grinda **JO** joista **DL** dusumea **PO** polita **C1/C2** coltar **CF** contrafisa

PIESE



DL

17

larice 28x145



H4

~204

2 / joista



F1

2

ulei

SCULE

Bormasina

Cui distantier 5 mm

Pensula

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

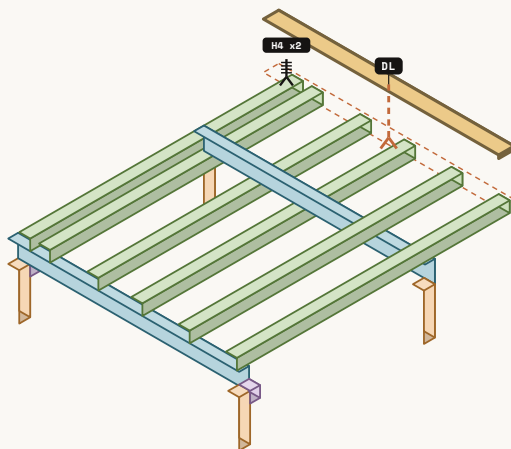
oblic

= la
unghi

Folosim **suruburi + buloane**, nu cuie.

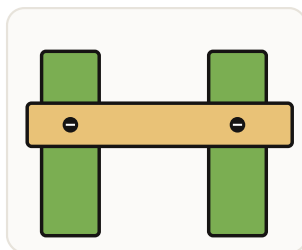
1

Aseaza prima scandura la marginea din fata, perpendicular pe joiste. 2 suruburi inox (H4) pe fiecare joista.

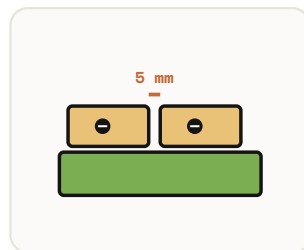


ZOOM · CUM SE PRINDE

DE SUS



LATERAL



H4 x2 / joista · INOX · gol de 5 mm

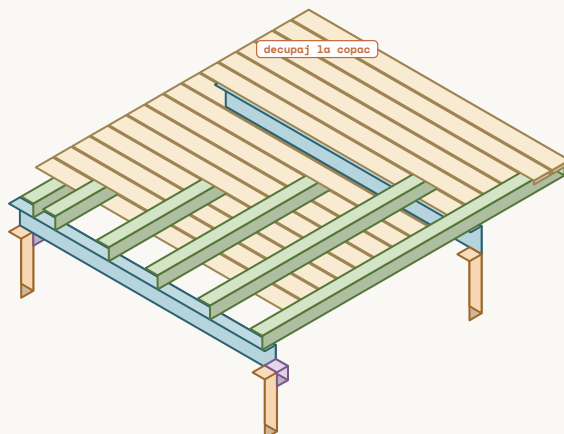
2

Lasa 5 mm intre scanduri (pune un cui ca distantier). Apa se scurge, lemnul respira.

! CRITIC — opreste-te si verifica.

3

Continua pana acoperi tot. La copac, taie scandurile in jurul gaurii, cu joc.



4

Dupa montaj, da cu ulei (F1) pe toata dusumeaua de larice.

GATA CAND

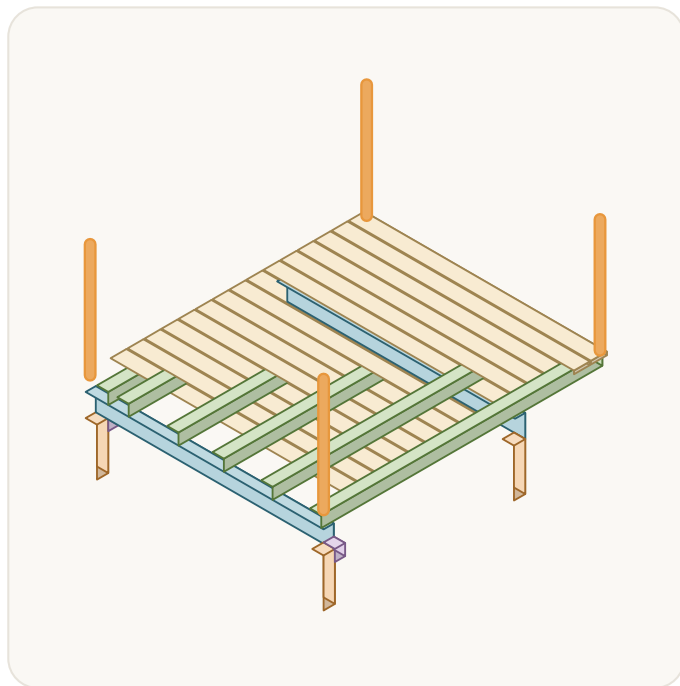
- ☐ Toate scandurile cu 2 H4 / joista
- ☐ Gol egal de 5 mm
- ☐ Decupaj la copac
- ☐ Uns cu ulei

FISA 11 / 11

Balustrada si poarta

~3-4 ore

CRITIC



ST stalp GR grinda JO joista DL dusumea PO polita C1/C2 coltar CF contrafisa

PIESE



SCULE

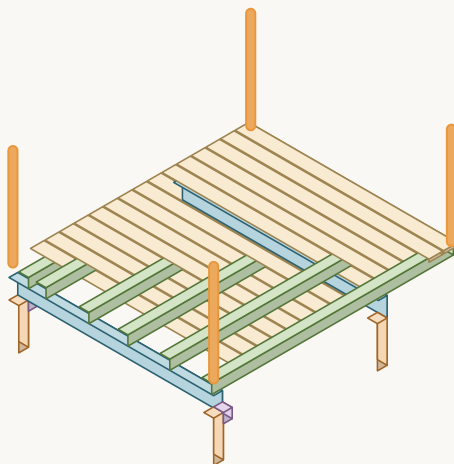
Bormasina Nivelata Fierastrau

PAS CU PAS

cap surub (fata) ascuns (pe spate) directie oblic = la unghi

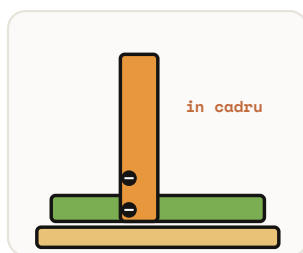
Folosim **suruburi + buloane**, nu cuie.

- 1** Monteaza stalpisorii balustradei, bulonati de CADRU (joiste/grinda), nu de dusumea. Jur-impregur, inclusiv pe nasul consolei.



ZOOM · CUM SE PRINDE

LATERAL



FATA

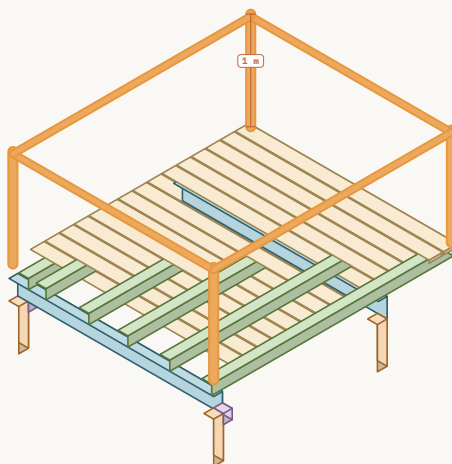


Stalpisor bulonat de CADRU · goluri sub 9 cm

! CRITIC — opreste-te si verifica.

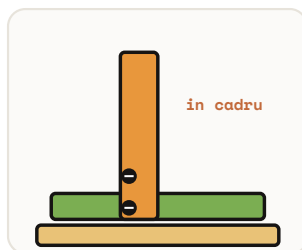
2

Pune mana curenta sus, la 1 m inaltime, pe tot conturul.

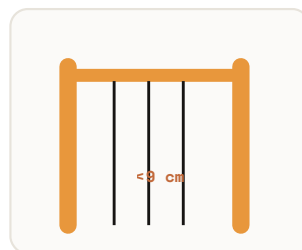


ZOOM • CUM SE PRINDE

LATERAL



FATA

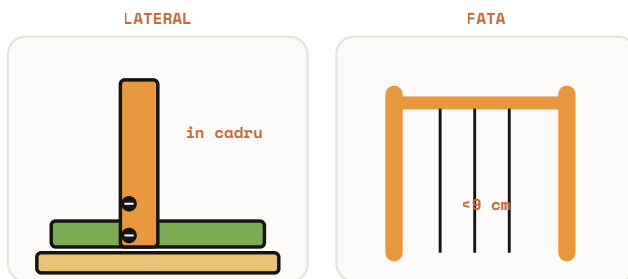


Stalpisor bulonat de CADRU • goluri sub 9 cm

! CRITIC — opreste-te si verifica.

3 Umple cu sipci, cu goluri sub 9 cm (un copil nu trebuie sa incapa sau sa se catere usor).

ZOOM · CUM SE PRINDE



Stalpisor bulonat de CADRU · goluri sub 9 cm

! **CRITIC** — opreste-te si verifica.

4 Lasa o poarta pe partea S3 (opusa mesei-copac), unde urca scara. Testeaza balustrada impingand cu putere.

! **CRITIC** — opreste-te si verifica.

GATA CAND

- ☐ Mana curenta la 1 m de la dusumea
- ☐ Goluri sub 9 cm peste tot
- ☐ Poarta pe S3
- ☐ Stalpisori bulonati de cadru, testati prin impingere

Siguranta & verificare

Cinci reguli care nu se negociaza — un copil va sta la 2,2 m.

Inaltime 2,2 m

Balustrada de 1 m jur-impresjur, goluri sub 9 cm, poarta pe partea S3. Sub platforma, sol moale (scoarta/nisip) pe ~5 m².

Copacul nu tine nimic

Corcodusul trece liber prin gaura cu joc de 3-5 cm. Blatul mesei se prinde de cadrul propriu, NICIODATA de copac.

Ordinea de siguranta

Proptelele raman pana montezi contravantuirile. Nimeni nu sta pe consola (balcon) inainte de contrafisa de sub nas.

Legaturi verificate

Coltare anti-ridicare la fiecare joista; coltar anti-smulgere la grinda din spate. Dupa montaj, reverifica strangerea buloanelor portante (polite + baza).

Lemn protejat

Grunduieste lemnul de structura INAINTE de montaj; ulei pe dusumea DUPA montaj. Capetele de stalp in papuc, cu drenaj, sa nu putrezeasca.